

Fig. 3: RS232 data format

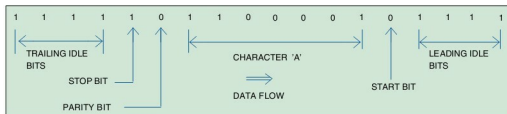


Fig. 4: Data frame when character 'A' is sent

standard since then. The most recent version of the standard is RS232 E and it is now addressed as EIA/TIA-232 E (Electronic Industry Association and Telecommunications Industry Association).

The RS232 standard specifies that a logic '1' is sent as a voltage in the range of -15V to -5V and logic '0' is sent as a voltage in the range of +5V to +15V. The standard specifies that a voltage of at least 3V amplitude will be recognised correctly at the receiver according to their polarity. It is an asynchronous standard and the transmitted bit sequence comprises a start bit, seven or eight data bits, an optional parity bit and one or more stop bits. The start bit is always low (logic '0' or space) and the stop bit is always high (logic '1' or mark). The parity bit used can be even parity or odd parity. The data format used for RS232 transmission is shown in Fig. 3. Fig. 4 shows the data frame used to send character 'A' when one stop bit and odd parity is used.

RS232 communication is implemented using a 9-pin connector (DB9) or a 25-pin connector (DB25). The DB25 connector is not used for normal communications as most of the pins are not needed. DB9 connector is used for serial communications using PCs. Figs 5 and 6 show the pin details of the DB25 and DB9 connectors, respectively.

The functionalities offered by the different pins of the DB9 connector are as follows:

TXD—This pin carries data from

the computer (DTE) to the serial device (DCE)

RXD—This pin carries data from the serial device (DCE) to the computer (DTE)

DTR—It is an indicator from DTE and is used by it to signal that it is ready to communicate with DCE

DSR—It is an indicator from DCE that it is on

RTS—Requests clearance to send data to DCE

CTS—Used by the DCE to acknowledge DTE's RTS signal

DCD—It is an indicator that DCE is connected to the telephone line

RI—It is an indicator that DCE has detected an incoming ring signal on the telephone line

RS232 was designed to be used for communication between a DTE (computer) and a DCE (modem).

In addition to communication between computer and a modem, RS232 is now widely used for direct connections between data acquisition devices and computer systems. In such cases, both the devices function as DTE. For this purpose, a null modem connection is used rather than having pin-to-pin connections of the modem cable. In a null modem connection, the transmission and the receiving lines are cross-linked. In some cases, the handshaking signals are also cross-linked, depending upon the application.

In a nutshell, null modem cables are used when two DTEs have to communicate with each other and pin-to-

pin connections are used when a DTE and a DCE have to communicate with each other.

Most systems designed today do not operate using RS232 voltage levels.

Therefore level conversion is necessary to implement

RS232 communication. Level conversion is performed using line drivers and receivers. Some of the line drivers/receivers used for performing translation include MAX232 (which is a dual EIA-232 driver/receiver intended for EIA/TIA 232 E and V.28 communication interfaces), MAX235 (which is a +5V powered multichannel EIA-232 driver/receiver also intended for EIA/TIA 232 E and V.28 communication interfaces, particularly suited to applications where $\pm 12V$ is not available) and so on.

VHDL code description

The VHDL code (rs232_fpga.vhd) of this article included in this month's EFY DVD is used to transmit and receive data using RS232 protocol at 19.2kbps using odd parity.

The code implements the transmitter and the receiver finite state machine (FSM) to transmit and receive data using RS232 protocol. The code begins with the standard library to be included. The Entity section defines the input and the output connections including the 50MHz clock (CLK), reset (RST), data byte to be transmitted (Din), latch data (LD) and serial data in (Rx) as inputs and data byte received (Dout), serial data out (Tx), transmit busy flag (TxBusy), receive error flag (RxErr) and receive ready flag (RxRDY) as outputs.

The different constants and signals used to implement the transmitter and the receiver FSMs are defined in the Architecture section. Constant Fxtal is an integer with a value of 50,000,000